



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE

PRN : 25070122055

Batch : 2025-29

Assignment : 15

Subject: Programming with Java

ASSIGNMENT No : 15

TITLE : Write a Java program to demonstrate exception handling using finally, throw, and throws.

Problem Statement

Write a Java program to demonstrate exception handling using finally, throw, and throws.

Learning Objectives

To implement finally, throw, and throws for effective exception management.

Theory

The finally block always executes regardless of whether an exception occurs, making it suitable for resource cleanup. The throw keyword explicitly generates an exception, while throws declares exceptions that a method may pass to its caller. These mechanisms improve structured error handling and application stability.



Code and Output Screenshots

```
J Nestedtry.java > Language Support for Java(TM) by Red Hat > Nestedtry
1 public class Nestedtry {
    Run | Debug | Run main | Debug main
2     public static void main(String[] args) {
3         try {
4             System.out.println("Outer block started.");
5
6             try {
7                 System.out.println("Inner try block started.");
8                 int result = 90/0;
9             } catch (ArithmeticException e) {
10                System.out.println("Caught arithmetic exception in inner try block" + e.getMessage());
11            }
12            String str = null;
13            System.out.println(str.length());
14
15        } catch (NullPointerException e) {
16            System.out.println("Caught null pointer exception in outer try block : " + e.getMessage());
17        } finally{
18            System.out.println("Finally block of outer try executed.");
19        }
20        System.out.println("Program continues after nested try catch blocks.");
21    }
22 }
23
24
```

PROBLEMS 39 OUTPUT DEBUG CONSOLE TERMINAL PORTS Code + v

```
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac Nestedtry.java }
Nestedtry.class
● Outer block started.
  Inner try block started.
  Caught arithmetic exception in inner try block/ by zero
  Caught null pointer exception in outer try block : Cannot invoke "String.length()" because "<local1>" is null
  Finally block of outer try executed.
  Program continues after nested try catch blocks.
○ PS C:\Users\LENOVO\Documents\JAVA Assignments>
```



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```
J EXP.java > ...
1 class InvalidAgeException extends Exception{
2     public InvalidAgeException(String m) {
3         super(m);
4     }
5 }
6 public class EXCP {
7     public static void validate(int age)
8         throws InvalidAgeException {
9         if(age<18) {
10             throw new InvalidAgeException("Age must be above 18.");
11         }
12         System.out.println("Valid age : " + age);
13     }
14     public static void main(String[] args) {
15         try {
16             validate(13);
17         } catch(InvalidAgeException e) {
18             System.out.println("Caught Exception : " + e.getMessage());
19         }
20     }
21 }
22
```

Run | Debug | Run main | Debug main

PROBLEMS 39 OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

```
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac EXCP.java }

Caught Exception : Age must be above 18.
PS C:\Users\LENOVO\Documents\JAVA Assignments>
```

```
J EXP.java > Language Support for Java(TM) by Red Hat > EXP
1 public class EXP {
2
3     Run | Debug | Run main | Debug main
4     public static void main(String[] args) {
5         try {
6             int[] n = new int[5];
7             int divisor = 0;
8
9             for (int i = 0; i < n.length; i++) {
10                 int res = n[i] / divisor;
11                 System.out.println(res);
12             }
13         } catch (ArithmeticException e) {
14             throw new RuntimeException("Error: Division by zero occurred", e);
15         }
16     }
17 }
```

PROBLEMS 39 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) {
Exception in thread "main" java.lang.RuntimeException: Error: Division by zero occurred
    at EXP.main(EXP.java:13)
    Caused by: java.lang.ArithmeticException: / by zero
    at EXP.main(EXP.java:9)
PS C:\Users\LENOVO\Documents\JAVA Assignments>
```



Conclusion : This program demonstrated the use of finally, throw, and throws in Java exception handling. The throw keyword was used to generate an exception, throws was used to declare it, and the finally block ensured that important code executed regardless of whether an exception occurred. Overall, exception handling improves program reliability, readability, and proper error management.

Exercise

Q1) Create a Login program that throws an exception for invalid password and uses finally block.

```
J pswd.java > Java > pswd
1  import java.util.Scanner;
2  public class pswd {
    Run | Debug | Run main | Debug main
3      public static void main(String[] args) {
4          Scanner sc = new Scanner(System.in);
5          String pass = "atharva123";
6          try {
7              System.out.print("Enter password: ");
8              String ps = sc.next();
9
10             if (!pass.equals(ps)) {
11                 throw new Exception("Invalid password");
12             }
13             System.out.println("Password is correct and valid.");
14         }
15         catch (Exception e) {
16             System.out.println(e.getMessage());
17         }
18         finally {
19             System.out.println("Password validator terminated.");
20             sc.close();
21         }
22     }
23 }
```

PROBLEMS 43 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?)
Enter password: atharva1
Invalid password
Password validator terminated.
○ PS C:\Users\LENOVO\Documents\JAVA Assignments> |
```



Ans) Case 1

Case 2

```
J pswd.java > Java > pswd
1  import java.util.Scanner;
2  public class pswd {
    Run | Debug | Run main | Debug main
3      public static void main(String[] args) {
4          Scanner sc = new Scanner(System.in);
5          String pass = "atharva123";
6          try {
7              System.out.print("Enter password: ");
8              String ps = sc.next();
9
10             if (!pass.equals(ps)) {
11                 throw new Exception("Invalid password");
12             }
13             System.out.println("Password is correct and valid.");
14         }
15         catch (Exception e) {
16             System.out.println(e.getMessage());
17         }
18         finally {
19             System.out.println("Password validator terminated.");
20             sc.close();
21         }
22     }
23 }
```

PROBLEMS 43 OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

```
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac pswd.java }

• Enter password: atharva123
  Password is correct and valid.
  Password validator terminated.
○ PS C:\Users\LENOVO\Documents\JAVA Assignments> 
```



Q2) Create an ATM PIN Verification program that throws an exception for an invalid PIN entered by the user and uses a finally block to display a message indicating that the verification process has completed.

Ans) Case 1

```
Pin.java > ...
1  import java.util.Scanner;
2  public class Pin {
    Run | Debug | Run main | Debug main
3      public static void main(String[] args){
4          Scanner sc = new Scanner(System.in);
5          int pno = 305609;
6          try{
7              System.out.println("Enter ATM pin to access bank account : ");
8              int pnoGuess = sc.nextInt();
9              if(pno != pnoGuess) {
10                 throw new Exception("Invalid pin number.");
11             }
12             System.out.println("Correct pin. Please proceed with further operations.");
13         }
14         catch (Exception e) {
15             System.out.println(e.getMessage());
16         }
17         finally{
18             System.out.println("PIN verification process completed.");
19         }
20     }
21 }
22
```

PROBLEMS 40 OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

```
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac Pin.java }
Enter ATM pin to access bank account :
1234
Invalid pin number.
PIN verification process completed.
PS C:\Users\LENOVO\Documents\JAVA Assignments>
```



Case 2

```
Pin.java > ...
1  import java.util.Scanner;
2  public class Pin {
    Run | Debug | Run main | Debug main
3      public static void main(String[] args){
4          Scanner sc = new Scanner(System.in);
5          int pno = 305609;
6          try{
7              System.out.println("Enter ATM pin to access bank account : ");
8              int pnoGuess = sc.nextInt();
9              if(pno != pnoGuess) {
10                 throw new Exception("Invalid pin number.");
11             }
12             System.out.println("Correct pin. Please proceed with further operations.");
13         }
14         catch (Exception e) {
15             System.out.println(e.getMessage());
16         }
17         finally{
18             System.out.println("PIN verification process completed.");
19         }
20     }
21 }
22
```

PROBLEMS 40 OUTPUT DEBUG CONSOLE TERMINAL PORTS Code

```
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac Pin.java }
Enter ATM pin to access bank account :
305609
Correct pin. Please proceed with further operations.
PIN verification process completed.
PS C:\Users\LENOVO\Documents\JAVA Assignments> |
```



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